Computer Networks

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Computer Networks

Computer Networks

- A **computer network** is a group of interconnected computer systems and devices that communicate and share resources through communication channels.
- In information technology, a computer network, also called a data network, is a series of points, or nodes, interconnected by communication paths for the purpose of transmitting, receiving and exchanging data, voice and video traffic.
- Computer Networking is simply a group of computers that are connected to each other in some way for the purpose of communicating to each other.
- **Uses:**
  - File sharing and resource sharing
  - Communication tools (e.g., voice and video)
  - Distributed systems (e.g., railway reservation, distributed databases)

Data Communications

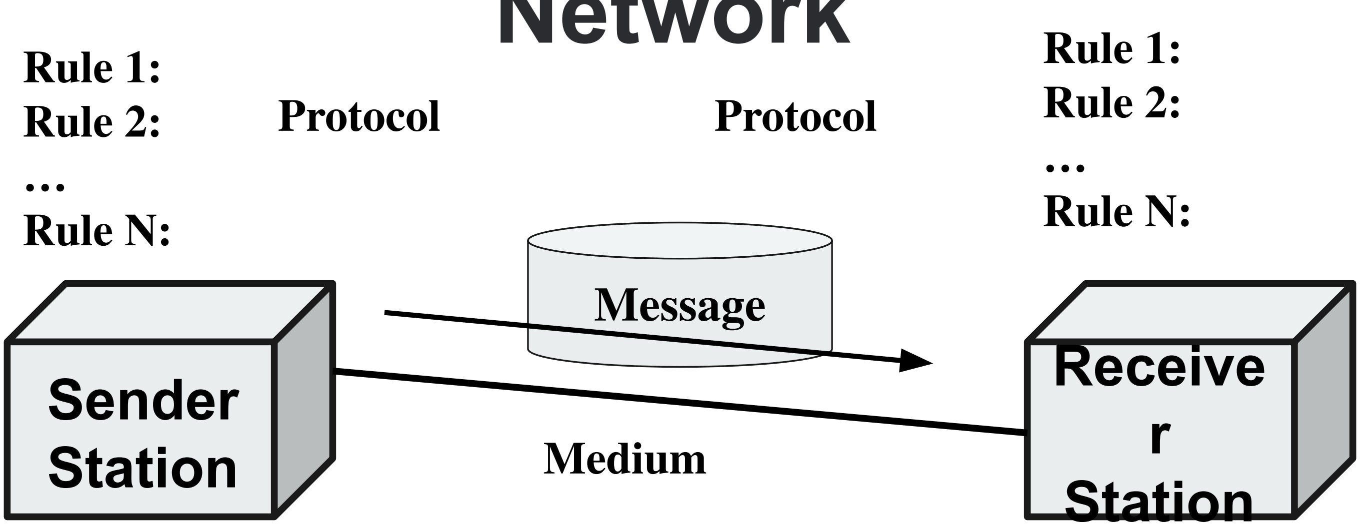
- Data communications is the exchange of data between two devices via some form of transmission medium such as a wire cable.
- For data communications to occur, the communicating devices must be part of a communications system made up of a combination of hardware (physical equipment) and software (programs).
- The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

Characteristics of Computer Network

- **Delivery:**
  - The system must deliver data to the correct destination.
- **Accuracy:**
  - The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
- **Timeliness:**
  - The system must deliver data in a timely manner. Data delivered late are useless.
- **Jitter:**
  - Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

Components of Computer Network

- **A data communications system has five components**



Components of Computer Network

- **Message:** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio etc.
- **Sender:** The sender is the device that sends the data message. It can be a computer, a telephone handset, a video camera, and so on.
- **Receiver:** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
- **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. Examples: twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
- **Protocol:** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not able to communicate, just as a person speaking.

Adv. of Computer Networks

- **Enhanced Communication**: Enables instant messaging and real-time communication.
- **Convenient Resource Sharing**: Easy sharing of files, printers, and other resources.
- **File Sharing**: Simplifies file sharing across systems.
- **Flexibility**: Users can access and share various information and software.
- **Cost-effective**: Lower cost for installation and resource sharing.
- **Increased Storage**: Allows efficient storage and management of data.

Disadv. Computer Networks

- ➢ **Lack of Independence**: Heavy reliance on computers and servers.
- ➢ **Security Risks**: Vulnerable to data breaches and unauthorized access.
- ➢ **Lack of Robustness**: System failure (e.g., server crash) can halt the entire network.
- ➢ **Viruses & Malware**: Increased risk of computer viruses and malicious attacks.
- ➢ **Negative Usage**: Network access may lead to distractions and misuse (e.g., gaming, browsing).
- ➢ **Requires Skilled Management**: Demands expertise for setup, configuration, and troubleshooting.
- ➢ **Expensive Setup**: Initial setup costs can be high for hardware and infrastructure.

Mode of Communication in Network

- ➢ A **mode of communication** refers to the way data is exchanged between the client and server in a client-server architecture.
- ➢ It determines how the client and server interact and synchronize their operations during communication.
- ➢ Modes of communication are crucial in defining the behavior, efficiency, and responsiveness of the system.
- ➢ Mode of Communications:
  - Simplex
  - Duplex
    - Half-Duplex
    - Full-Duplex

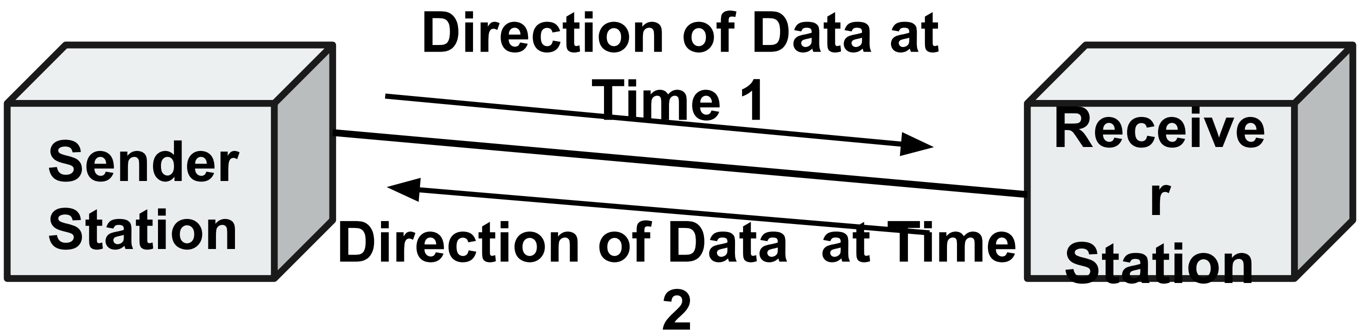
Simplex

Simplex

- ➢ The simplest signal flow technique is the simplex configuration.
- ➢ In Simplex transmission, one of the communicating devices can only send data, whereas the other can only receive it.
- ➢ Communication/Transmission is only in one direction (unidirectional) where one party is the transmitter and the other is the receiver.

Examples of simplex communication are the simple radio, and Public broadcast television where, you can receive data from stations but can't transmit data back. The television station sends out electromagnetic signals. The station does not expect and does not monitor for a return signal from the television set. This type of channel design is easy and inexpensive to set up.

Half-Duplex

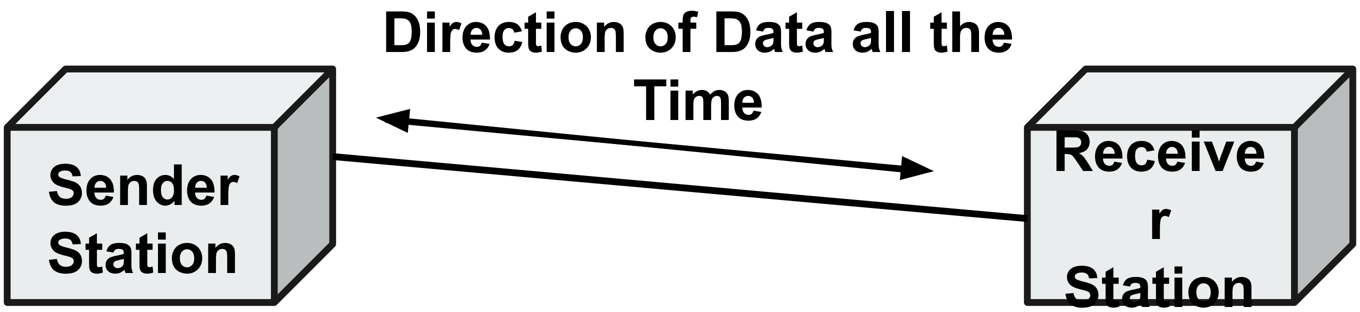


Half-Duplex

- Half duplex refers to two-way communication where, only one party can transmit data at a time.
- Both devices (Sender and Receiver) can transmit data though, not at the same time, that is Half duplex provides Simplex communication in both directions in a single channel. When one device is sending data, the other device must only receive it and vice versa.
- This requires a definite turnaround time during which, the device changes from the receiving mode to the transmitting mode. Due to this delay, half duplex communication is slower than simplex communication. However, it is more convenient than simplex communication as both the devices can send and receive data.

  For example, a walkie-talkie is a half-duplex device because only one party can talk at a time.

Full-Duplex



Full-Duplex

- Full duplex refers to the transmission of data in two directions simultaneously.
- Both the devices are capable of sending as well as receiving data at the same time.
- Sharing the same channel and moving signals in both directions increases the channel throughput without increasing its bandwidth.

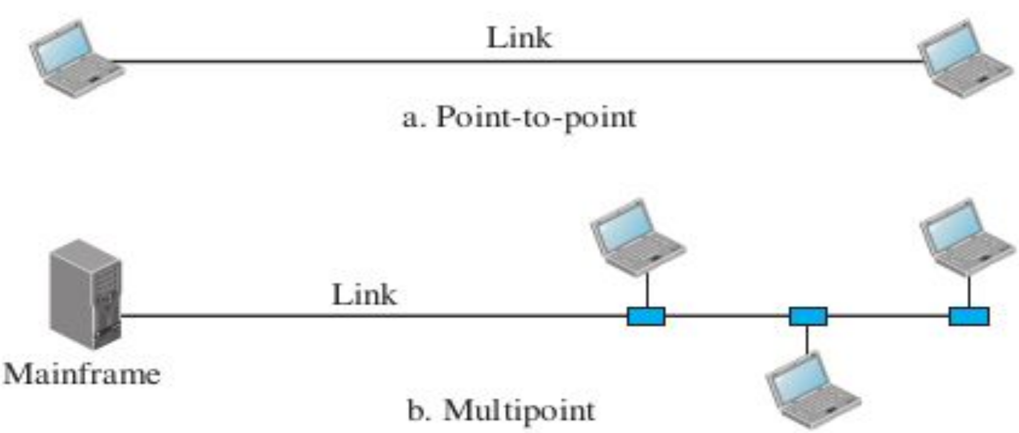
For example, a telephone is a full-duplex device because both parties can talk to each other simultaneously.

Network Criteria

- **Performance:** Performance can be measured in many ways, including transit time and response time.
  - Transit time is the amount of time required for a message to travel from one device to another.
  - Response time is the elapsed time between an inquiry and a response.
- **Reliability:** Network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.
- **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Types of Connection

- A network is two or more devices connected through links.
- A link is a communications pathway that transfers data from one device to another.
- There are two possible types of connections: Point-to-point and Multipoint.



Types of Connection

- **Point-to-point**
  - A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices.
- **Multipoint**
  - A multipoint (also called multidrop) connection is one in which more than two devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is as spatially shared connection. If users must take turns, it is a timeshared connection.

Computer Networks Models

- Based on Scale(Size)
  - PAN, LAN, WAN, MAN
- Based on Topology
  - Physical Topology
    - Ring, Bus, Star, Mesh, Hierarchical, Hybrid Topology
  - Logical Topology
- Based on Architecture
  - Peer-to-Peer Architecture
  - Client-Server Architecture
    - One-Tier, Two-Tier, Three-Tier, N-Tier Architecture
    - Peer-to-Peer, Thin-Client, Fat-Client Architecture

Computer Networks Models

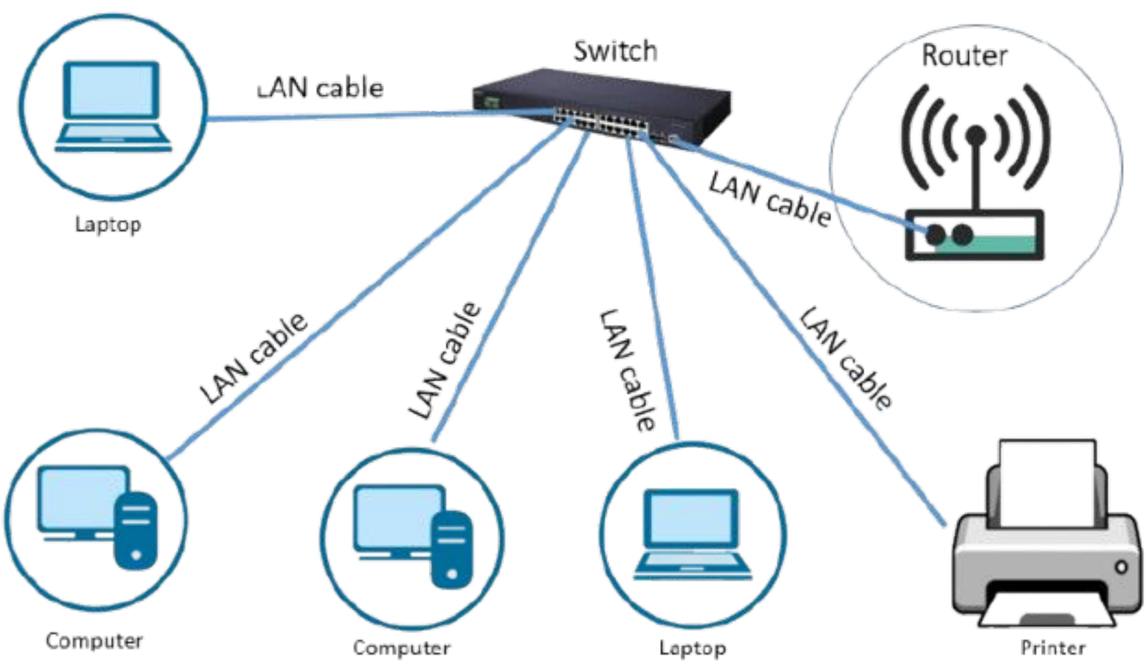
- Based on Scale(Size)
  - PAN - Personal Area Networks
  - LAN - Local Area Networks
  - WAN - Wide Area Networks
  - MAN - Metropolitan Area Networks

PAN: Personal Area Network

PAN: Personal Area Network

- **Personal Area Network (PAN)** is a small-scale network used for data transmission between personal devices like computers, mobile phones, PDAs, and other electronic gadgets.
- **Key Characteristics of PAN:**
  - **Limited Range**: It can covers a short distance, usually within a range of up to 10 meters.
  - **Medium**: Common technologies used for PANs include **Bluetooth** and **Infrared**.
  - **Few Connections**: A PAN usually supports a small number of devices, typically 2 to 8.
- **Applications of PAN:**
  - **Device Connectivity**: Connecting devices like smartphones, laptops, tablets, and wireless headphones.
  - **Data Sharing**: Allowing quick file transfers and device synchronization over short distances.

LAN: Local Area Network



LAN: Local Area Network

- ➢ It refers to a network of interconnected devices that are under the same administrative control, typically within a limited geographic area, such as a home, office, or campus.
- ➢ Initially, LANs were defined as small networks in a single physical location, but today, they can encompass multiple interconnected networks across different buildings and locations.

- ➢ **Key Characteristics of LANs:**
  - **Limited Geographic Area**: LANs operate within a confined space, such as a home or office.
  - **High Bandwidth**: LANs provide multi-access to high-speed media for efficient data transfer.

LAN: Local Area Network

- ➢ **Components of a LAN:**
  - **Computers**: The devices that make up the network.
  - **Network Interface Cards (NICs)**: Hardware that connects each computer to the network.
  - **Peripheral Devices**: Devices like printers, scanners, etc., that are connected to the network.
  - **Networking Media**: The physical medium (e.g., cables or wireless signals) through which data is transmitted.
  - **Network Devices**: Equipment such as routers, switches, or hubs that manage and direct network traffic.

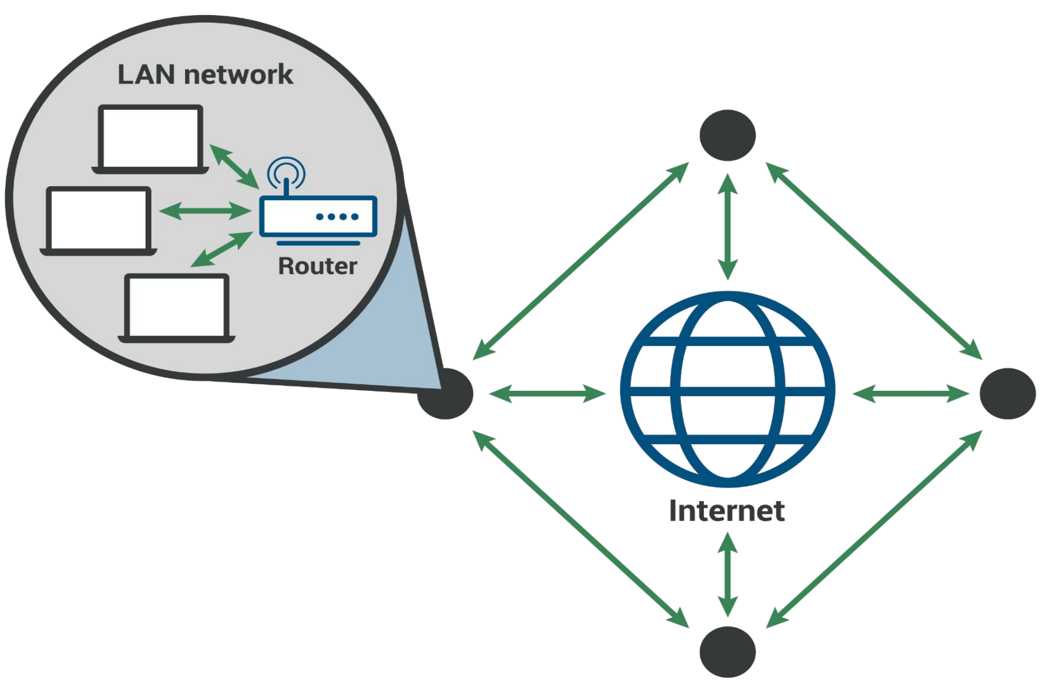
LAN: Local Area Network

- **Uses of LANs:**
  - **File and Printer Sharing**: LANs allow multiple computers to share resources like files and printers.
  - **Internal Communications**: LANs enable internal communication within an organization, e.g., email systems.

Common LAN Technologies:

  - **Ethernet**: A widely used LAN technology for high-speed data transmission.
  - **Token Ring**: An older LAN technology where devices take turns sending data in a specific sequence.

WAN: Wide Area Network



WAN: Wide Area Network

- It is a network that covers a broad geographic area, interconnecting multiple Local Area Networks (LANs) over long distances, often using public or leased communication infrastructure.
- WANs enable businesses, organizations, and individuals to communicate across vast distances and share resources like computers, file servers, and printers.
- **Key Characteristics of WANs:**
  - **Large Geographic Area**: WANs span cities, countries, or even continents, connecting networks over a vast area.
  - **Real-Time Communication**: WANs support real-time communication, such as voice, video, and instant messaging, allowing for seamless collaboration.
  - **Services**: WANs offer services like **email**, **Internet access**, **file transfer**, and **e-commerce**, which are essential for business operations across multiple locations

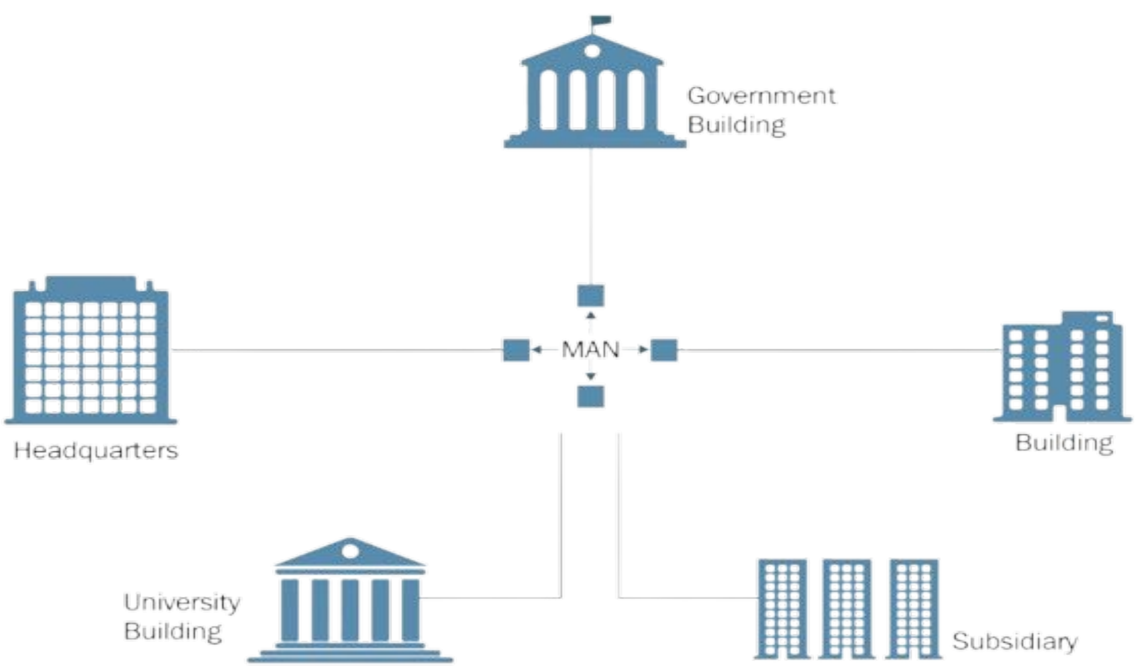
WAN: Wide Area Network

- **Common WAN Technologies:**
  - **Modems**: Devices that modulate and demodulate digital signals for transmission over telephone lines.
  - **Integrated Services Digital Network (ISDN)**: A digital network for voice and data transmission.
  - **Digital Subscriber Line (DSL)**: A high-speed internet connection using telephone lines.
  - **Frame Relay**: A data link layer protocol for connecting networks over WANs.
  - **T1, E1, T3, E3**: High-speed leased lines for transmitting data over large distances.
  - **Synchronous Optical Network (SONET)**: A fiber-optic standard used for high-speed WAN communication.

WAN: Wide Area Network

- **Applications of WAN:**
  - **Remote Work**: WANs enable telecommuters to access company resources and work from home, without needing to be physically present in the office.
  - **Global Communication**: WANs support global communication across business units, clients, and partners.
  - **Collaboration**: WANs enable collaboration through real-time information sharing, video conferences, and remote meetings.

MAN: Metropolitan Area Network



MAN: Metropolitan Area Network

- ➢ It is a data network designed to cover a larger geographic area than a Local Area Network (LAN) but smaller than a Wide Area Network (WAN).
- ➢ MANs typically span towns or cities and are used to connect multiple LANs within a metropolitan area.
- ➢ **Key Characteristics of MANs:**
  - **Geographic Coverage**: MANs typically cover areas up to 50 kilometers, such as towns, cities, or large campuses.
  - **High-Speed Connections**: They use high-speed connections, often employing **fiber-optic cables** or other high-performance digital media.
  - **Medium**: MANs often use **optical fiber** and **copper cables** to transmit data efficiently.
  - **Data Rates**: They offer sufficient bandwidth to support **distributed computing applications** and ensure reliable communication between LANs.

MAN: Metropolitan Area Network

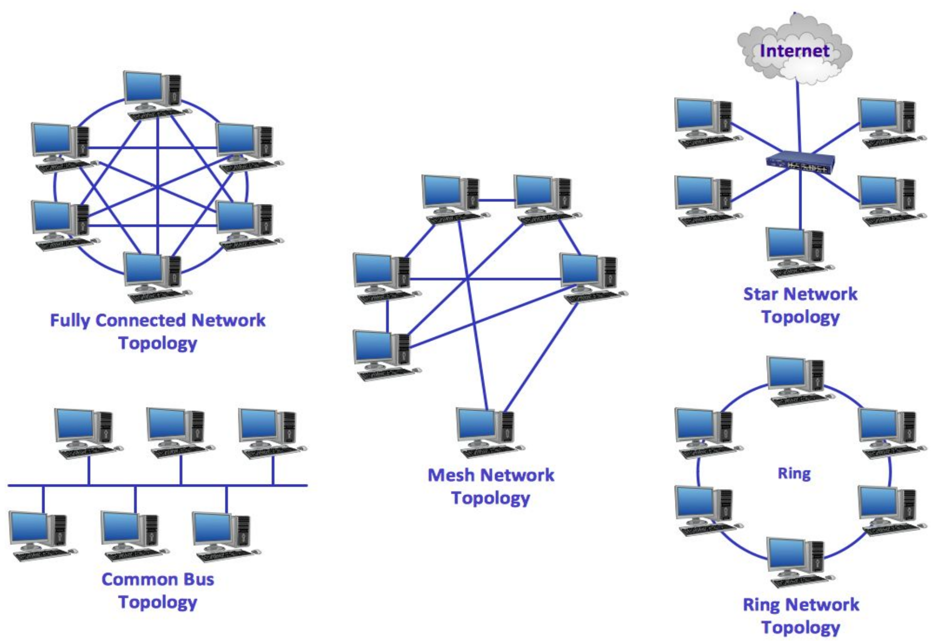
- ➢ **Comparison to LAN:**
  - **LAN**: Operates within a limited area, like a home or office, and connects devices within a smaller geographic scope (e.g., a building or campus).
  - **MAN**: Bridges the gap between LANs by connecting multiple LANs within a larger geographic area, like a city or town. It typically supports faster data transfer and larger user bases compared to LANs.
- ➢ **Applications of MAN:**
  - **Citywide Networks**: Connecting different parts of a city, such as local government offices, universities, and businesses.
  - **Internet Backbone**: MANs are often used as a backbone for regional Internet service providers (ISPs) to link various LANs across a metropolitan area.

Network Topology

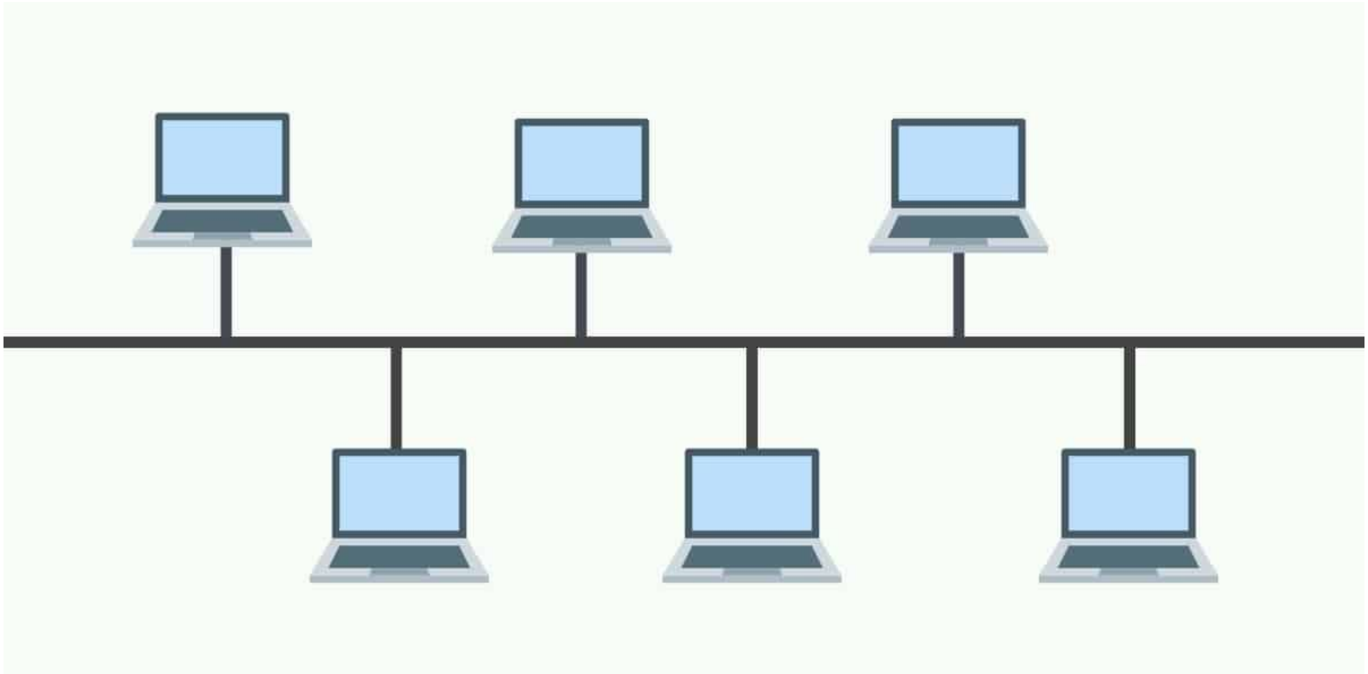
- ➢ Network topology is a topological structure of computer network, which can be physical or logical.
- ➢ **Physical topology:** The term physical topology refers to the way in which a network is laid out physically. The actual layout of the wire or media. Two or more devices connect to a link; two or more links form a topology.
- ➢ **Logical topology:** Defines how the hosts access the media to send data. Shows the flow of data on a network.

Computer Networks Models

- ➢ Based on Topology
  - ○ Physical Topology
    - ■ Ring Topology
    - ■ Bus Topology
    - ■ Star Topology
    - ■ Mesh Topology
    - ■ Hybrid Topology
  - ○ Logical Topology



Bus Topology



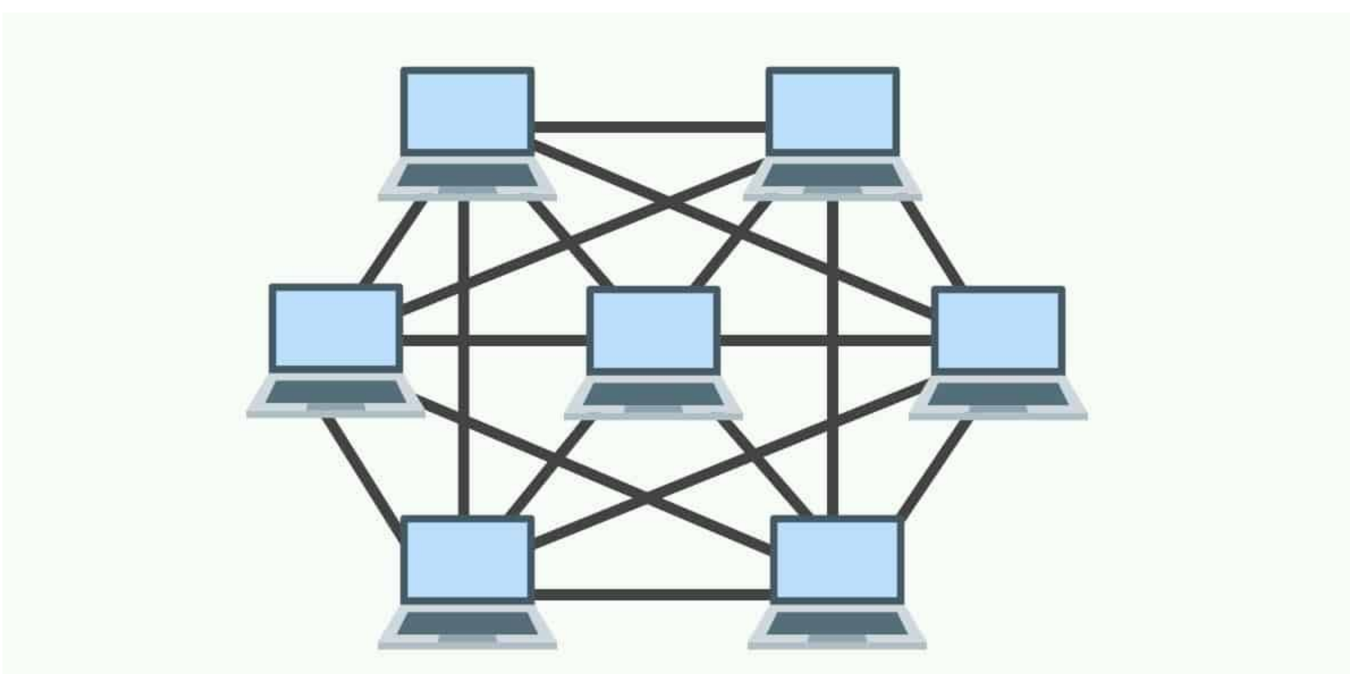
Bus Topology

- A networking topology that connects networking components along a single cable or that uses a series of cable segments that are connected linearly.
- A network that uses a bus topology is referred to as a “bus network.” Bus networks were the original form of Ethernet networks, using the 10Base5 cabling standard.
- **Bus topology is used for:**
    - Small work-group local area networks (LANs) whose computers are connected using a thinnet cable
    - Trunk cables connecting hubs or switches of departmental LANs to form a larger LAN
    - Backboning, by joining switches and routers to form campus-wide networks

Bus Topology

- **Advantages:**
  - Easy to install
  - Costs are usually low
  - Easy to add systems to network
  - Great for small networks
- **Disadvantages:**
  - Out of date technology.
  - Include difficult reconnection and fault isolation
  - Can be difficult to troubleshoot
  - Unmanageable in a large network
  - If cable breaks, whole network is down

Mesh Topology



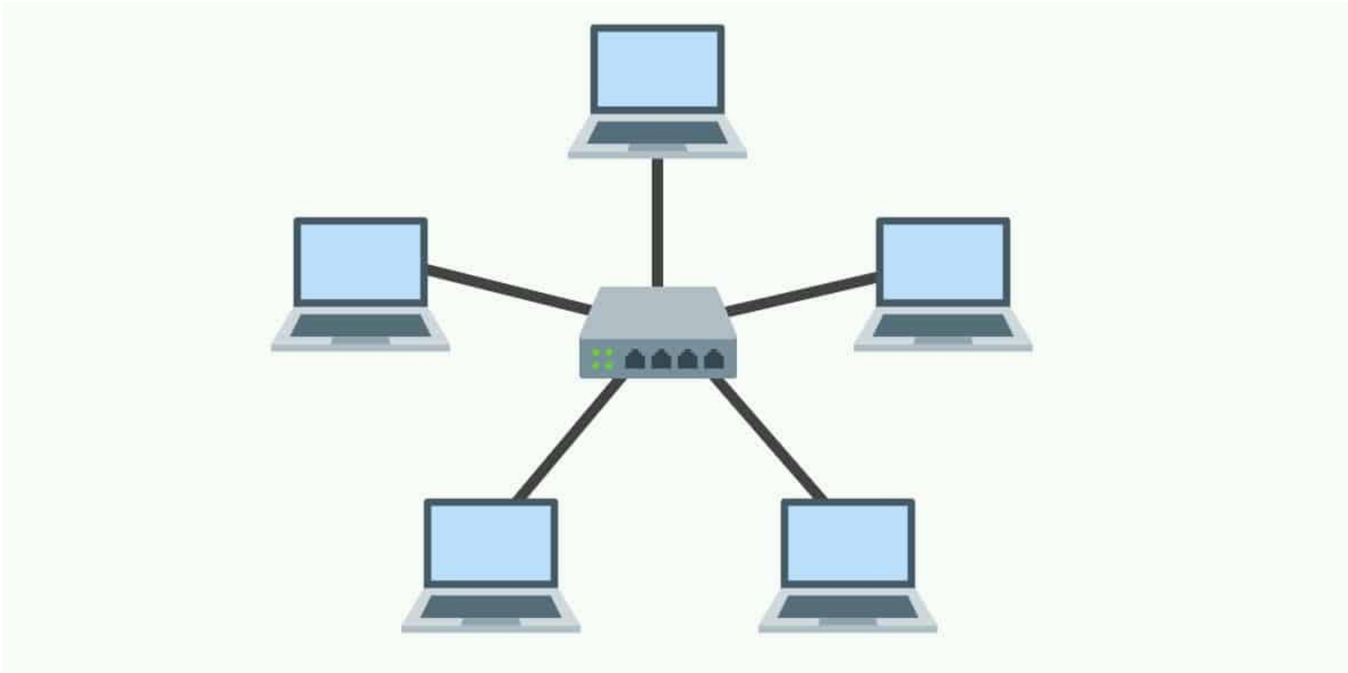
Mesh Topology

- In a mesh topology, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects. To connect n nodes in Mesh topology, we require n(n-1)/2 duplex mode links.
- One practical example of a mesh topology is the connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

Mesh Topology

- **Advantages:**
  - The use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
  - Robust, If one link becomes unusable, it does not incapacitate the entire system.
  - Advantage of privacy or security.
  - point-to-point links make fault identification and fault isolation easy , Traffic can be routed to avoid links with suspected problems.
- **Disadvantage:**
  - Required high amount of cabling and the number of I/O ports.
  - The sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
  - The hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Star Topology



Star Topology

- ➢ In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- ➢ The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- ➢ The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

Star Topology

➢ **Advantages:**
  ○ Less Expensive than Mesh topology.
  ○ In a star topology, each device needs only one link, which makes easy to install and re-configure.
  ○ Less Cabling, Addition and Deletion involves only one connection between the devices and the Hub or Switch.
  ○ Easy for Fault identification and fault isolation. If one link fails, only that link is affected.

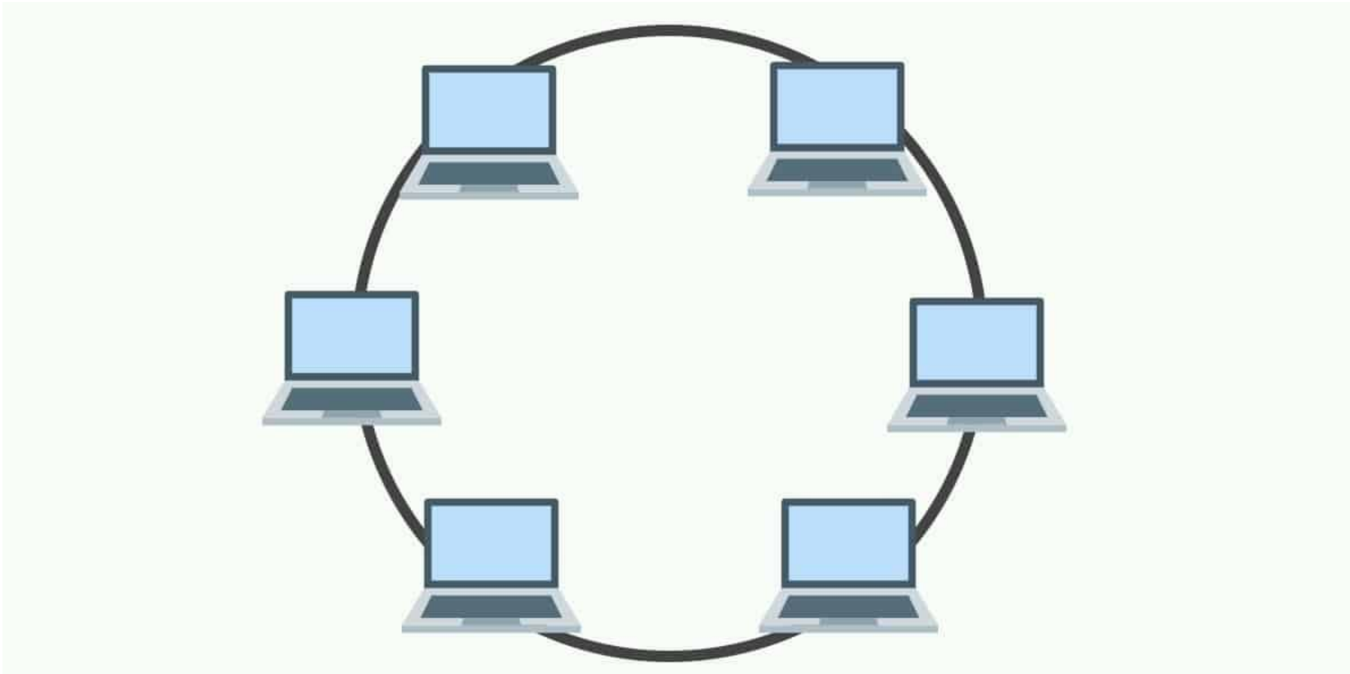
➢ **Disadvantage:**
  ○ One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.

Extended Star Topology

➢ An extended star topology links individual stars together by connecting the hubs or switches.

➢ A hierarchical topology is similar to an extended star. However, instead of linking the hubs or switches together, the system is linked to a computer that controls the traffic on the topology.

Ring Topology



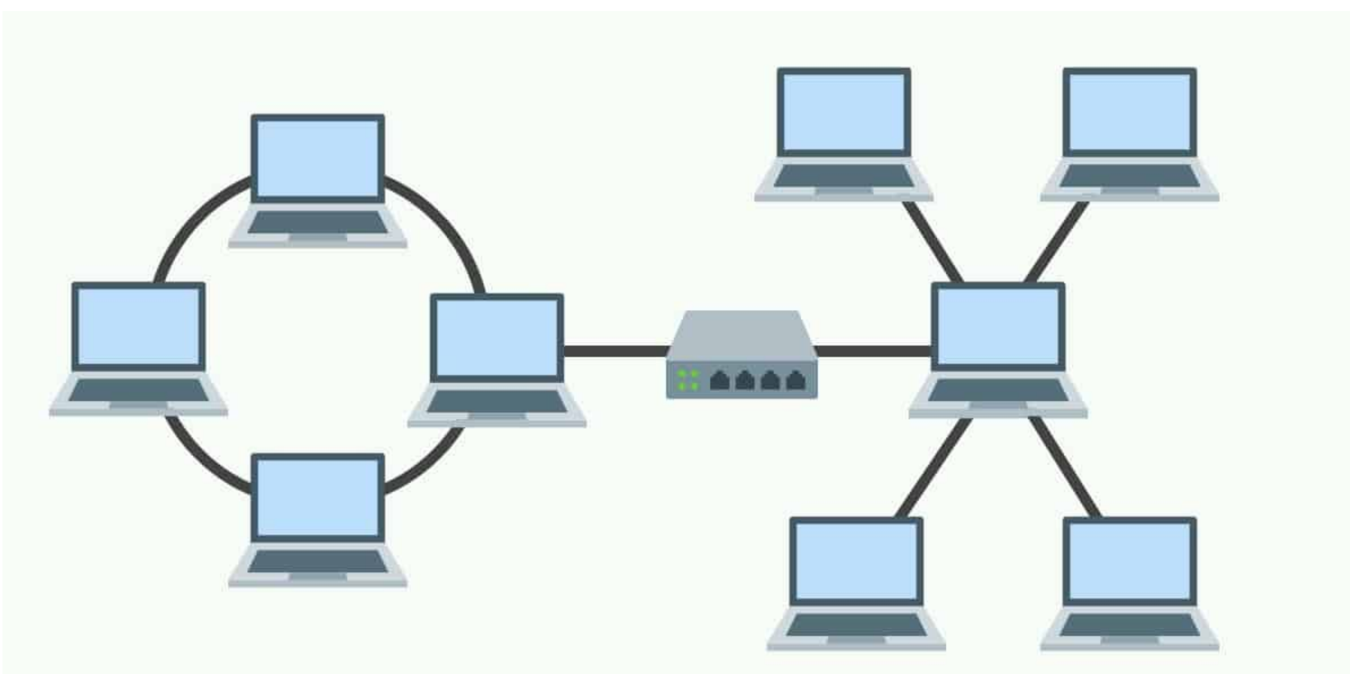
Ring Topology

- ➢ In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- ➢ Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.
- ➢ Each device is linked to only its immediate neighbors (either physically or logically).
- ➢ Generally in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

Ring Topology

- **Advantages:**
  - Very orderly network where every device has access to the token and the opportunity to transmit
  - Performs better than a bus topology under heavy network load
  - Does not require network server to manage the connectivity between the computers
- **Disadvantage:**
  - One malfunctioning workstation or bad port in the MAU can create problems for the entire network
  - Moves, adds and changes of devices can affect the network
  - Network adapter cards and MAU's a Multistation Access Unit are much more expensive than Ethernet cards and hubs
  - Much slower than an Ethernet network under normal load

Hybrid Topology



Hybrid Topology

- ➢ A hybrid network topology is a combination of two or more different types of topologies.
- ➢ For example, a network might combine star and mesh network topologies to leverage the strengths of each.
- ➢ This approach allows for customization to meet specific needs, balancing cost, performance, and reliability.

Logical Topology

- ➢ The logical topology of a network determines how the hosts communicate across the medium. The two most common types of logical topologies are **broadcast** and **token passing**.
  - The use of a **broadcast topology** indicates that each host sends its data to all other hosts on the network medium. There is no order that the stations must follow to use the network. It is first come, first serve. Ethernet works this way as will be explained later in the course.
  - In **token passing topology**, an electronic token is passed sequentially to each host. When a host receives the token, that host can send data on the network. If the host has no data to send, it passes the token to the next host and the process repeats itself.

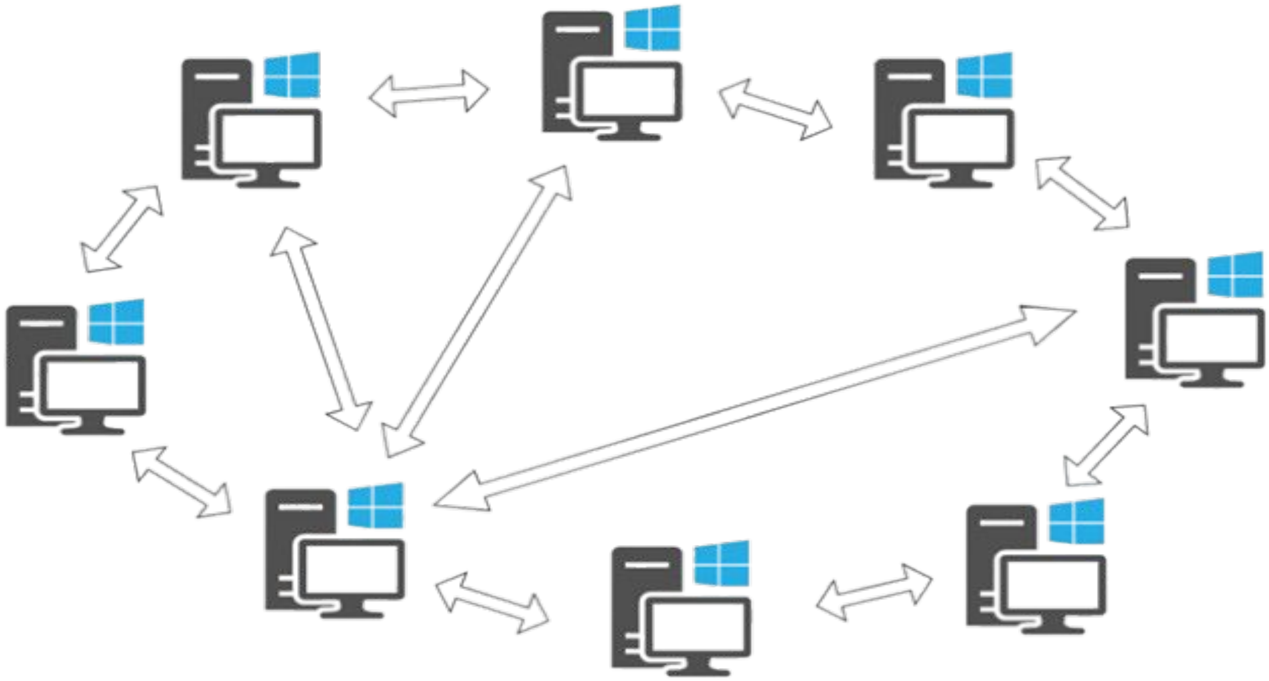
Computer Networks Models

- Based on Architecture
  - Peer-to-Peer Architecture
  - Client-Server Architecture
    - One-Tier, Two-Tier, Three-Tier, N-Tier Architecture
    - Peer-to-Peer, Thin-Client, Fat-Client Architecture

Network Architecture

- **Network Architecture** refers to the design and structure of a computer network.
- **Network Architecture** defines the layout of the network, how devices are interconnected, how data flows between devices, and how the network operates and is managed.
- **Network Architecture** includes the physical components, the protocols used, and the organizational structure of the network.
- **Network architecture** also encompasses the different layers that make up the network and their interactions.
- Key Elements of Network Architecture:
  - Topology, Protocols, Devices, Transmission Media,Security, Scalability

Peer-to-Peer Architecture



Peer-to-Peer Model/Architecture

- Networked computers act as equal partners, or peers. As peers, each computer can take on the client function or the server function.
- Individual users control their own resources. The users may decide to share certain files with other users. The users may also require passwords before they allow others to access their resources.
- Computer A may request for a file from Computer B, which then sends the file to Computer A. Computer A acts like the client and Computer B acts like the server. At a later time, Computers A and B can reverse roles.
- Since individual users make these decisions, there is no central point of control or administration in the network.

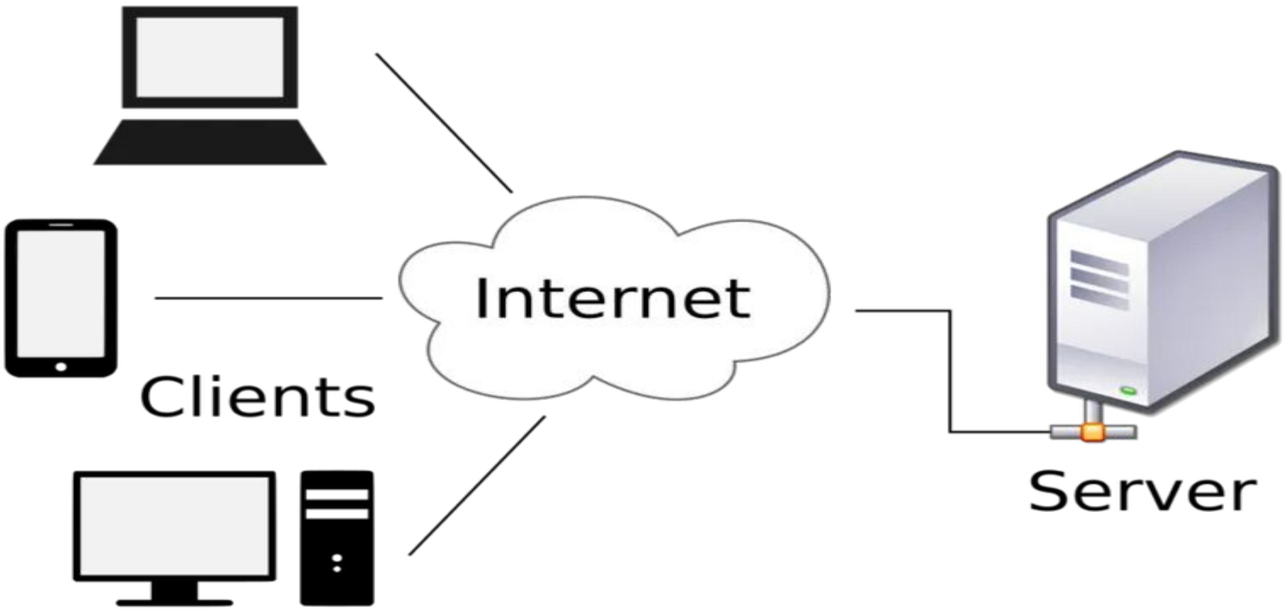
Adv. of Peer-to-Peer Networks

- **Simplicity and Ease of Setup**:
  - Easy to install and configure.
  - No need for additional equipment beyond a suitable operating system.
- **Cost-Effectiveness**:
  - No central server required, reducing hardware and maintenance costs.
- **Equal Resource Sharing**:
  - Each peer can share resources like files and printers without relying on a central server.
- **User Autonomy**:
  - Individual users control their own files and resources, deciding what to share and with whom.

Disadv. Of Peer-to-Peer Networks

- **Lack of Centralized Control**:
  - No central point of administration, making network coordination challenging.
- **Scalability Issues**:
  - Works efficiently with a small number of devices, and Performance and efficiency drop as the number of peers increases.
- **Reduced Performance**:
  - When a computer acts as a server, it may experience reduced performance due to serving requests from others.
- **Security Challenges**:
  - Security is difficult to maintain since individual users control access to their resources.
- **No Centralized Backup**:
  - Each user must back up their own data. Recovery from data loss is the user's responsibility.

Client-Server Architecture

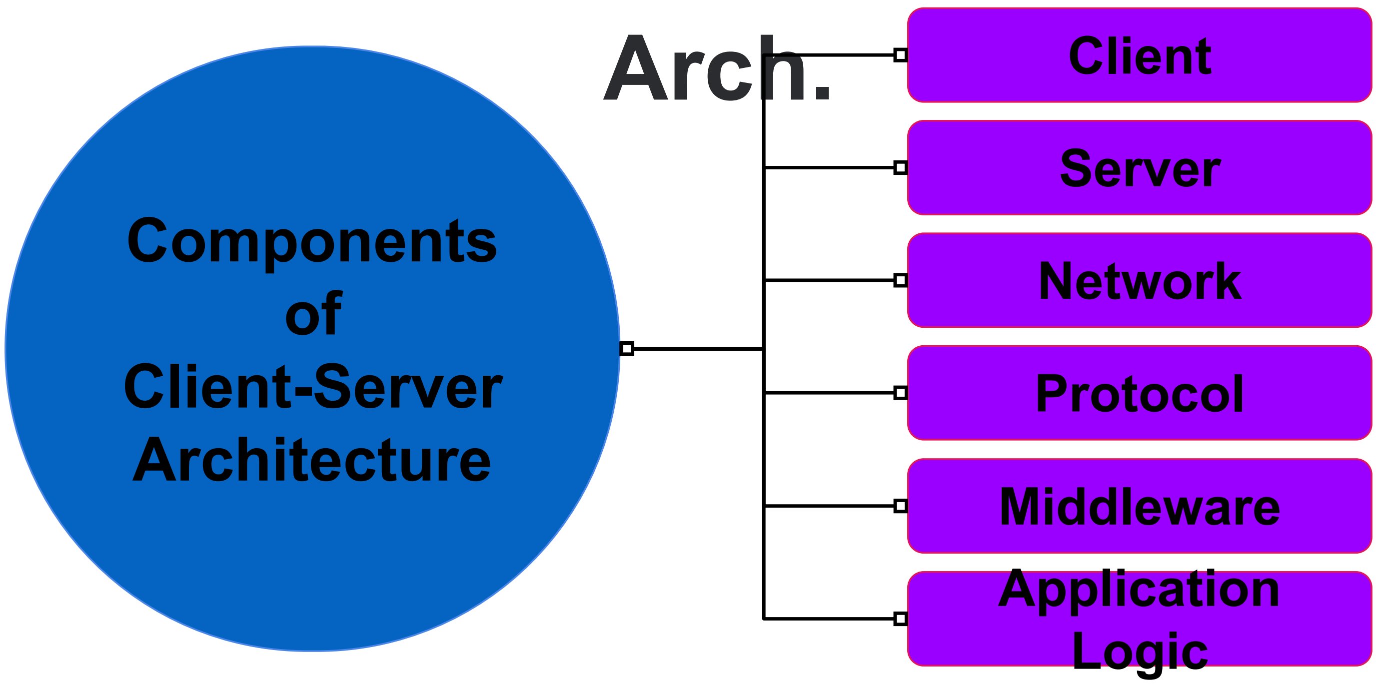


Client-Server Model/Architecture

- ➢ The term client-server refers to a popular model for computer networking that utilizes client and server devices each designed for specific purposes.
- ➢ The client-server model can be used on the Internet as well as local area networks (LANs).

    Examples of client-server systems on the Internet include Web browsers and Web servers, FTP clients and servers, and DNS.
- ➢ In a client/server arrangement, network services are located on a dedicated computer called a server. The server responds to the requests of clients. The server is a central computer that is continuously available to respond to requests from clients for file, print, application, and other services.
- ➢ Most network operating systems adopt the form of a client/server relationship.

Components of Client-Server Arch.



Components of Client-Server Arch.

- **Client:**
  - A device or application that requests services from a server.
  - Examples: Web browsers, mobile apps, and desktop applications.
- **Server:**
  - A computer or program that processes client requests and provides services like data storage, computations, or hosting.
  - Handles backend tasks such as business logic and data management.
- **Network:**
  - Connects clients and servers, facilitating data exchange.
  - Ranges from local area networks (LAN) to the Internet.

Components of Client-Server Arch.

- **Protocol:**
  - Defines rules for communication between clients and servers.
  - Examples: HTTP/HTTPS (web), FTP (file transfer), SMTP (email).
- **Middleware:**
  - Acts as a bridge between client and server, enabling communication.
  - Handles tasks like authentication, load balancing, and data translation.
- **Application Logic:**
  - Code that determines how the server processes and responds to client requests.
  - Includes business rules, data manipulation, and workflows.

Types of Client-Server Arch.

Client-Server Architecture

- **One-Tier Architecture**: All components (client, server, database) are on a single machine, Simple for small apps but lacks scalability and is rarely used in production.
- **Two-Tier Architecture**: Direct communication between client (UI and logic) and server (data storage), Easy to implement but limited in scalability, performance, and security.
- **Three-Tier Architecture**: Adds middleware for business logic between client (UI) and server (data storage), Improves scalability and security but increases complexity and cost.
- **N-Tier Architecture**: Extends three-tier with additional layers for modularity and flexibility, Suitable for complex systems but needs more resources and management.

Types of Client-Server Arch.

Client-Server Architecture

- **Peer-to-Peer Architecture**: Nodes act as both client and server, sharing resources without a central authority (e.g., BitTorrent). Improves reliability and resilience but faces challenges like security and coordination.
- **Thin-Client Architecture**: Minimal client functionality relies on the server for processing and storage (e.g., Google Docs). Reduces client-side costs but increases network dependency and server load.
- **Fat-Client Architecture**: The client handles most processing and storage locally, syncing with the server occasionally (e.g., Evernote). Enhances performance and availability but demands higher hardware and risks data inconsistency.

Comm. in Client-Server Arch.

- **Synchronous Communication:**
  - The client sends a request and waits for the server to respond before proceeding.
  - Ensures real-time interaction but may lead to delays if the server is busy.
- **Asynchronous Communication:**
  - The client sends a request and continues its tasks without waiting for the server's response.
  - Improves efficiency and performance but may complicate error handling and synchronization.

Adv. Client-Server Networks

- **Centralized Control:**
  - All resources are centrally managed, user authentication and access control.
- **Improved Security:**
  - Centralized management allows for better implementation of security policies.
- **Scalability:**
  - Servers can handle requests from multiple clients simultaneously, and easy to add or upgrade servers to meet growing demands.
- **Efficiency in Resource Utilization:**
  - Specialized servers optimize performance for specific tasks.
- **Ease of Maintenance and Backup:**
  - Centralized data makes it easier to perform backups and maintain systems.
- **Portability and Flexibility:**
  - Clients can be located anywhere and still access server resources

Disadv. Client-Server Networks

- **Single Point of Failure:**
  - If the central server fails, the entire network may become inoperable.
- **High Cost:**
  - Requires specialized hardware and software for servers.
  - Adds expenses for installation and maintenance.
- **Requires Skilled Staff:**
  - Servers need to be managed by trained professionals.
- **Complex Setup:**
  - Initial setup and configuration of a client-server network are more complex than peer-to-peer networks.
- **Dependency on Network Availability:**
  - Network interruptions can limit client access to server resources.
- **Performance Bottleneck:**
  - Servers may become overwhelmed with requests if not appropriately scaled.

Types of Computer Networks

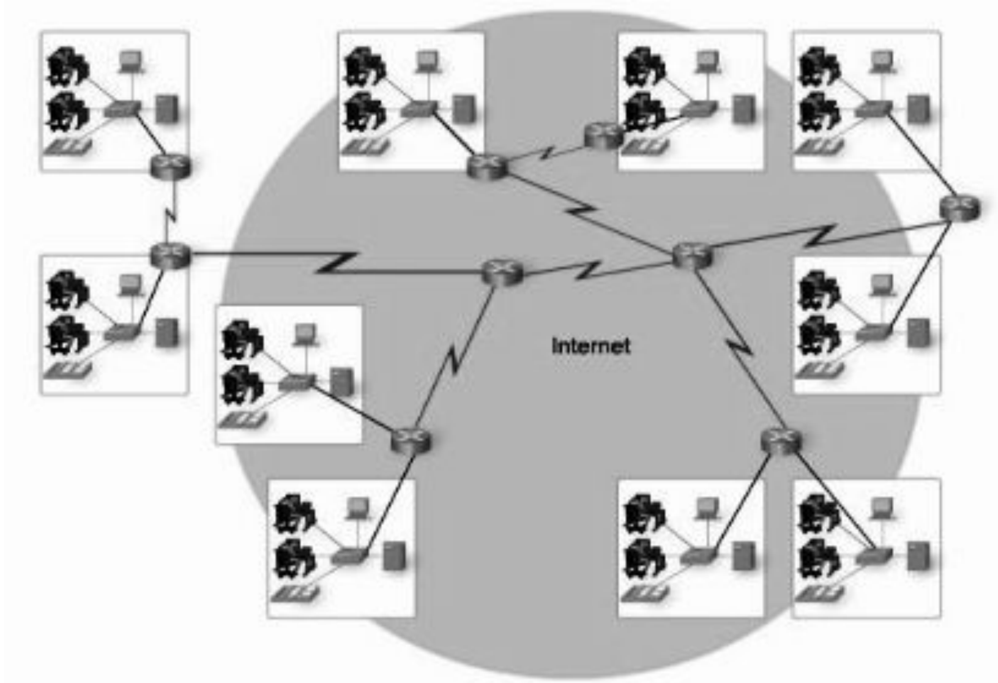
- Internet
- Intranet
- Extranet

Ethernet

Internet

- The **Internet** is a vast network formed by the cooperative interconnection of numerous smaller computer networks.
- It is often referred to as a **network of networks**, and it operates under the following key principles:
  - **No single ownership**: The Internet is not owned by any single entity; instead, each person or organization that connects to the Internet controls a portion of it.
  - **No central administration**: There is no central governing body managing the entire Internet; its structure is decentralized.

Internet



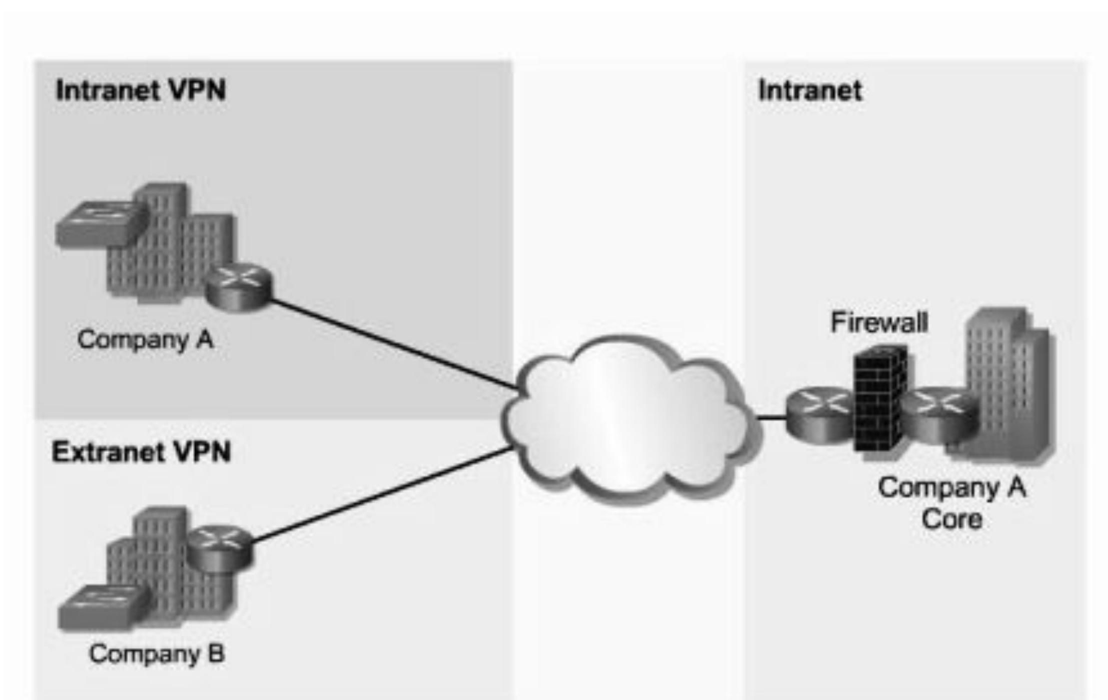
Internet

➢ The Internet consists of:
  - **A community of users**: People who utilize and contribute to the network by developing content, tools, and services.
  - **A collection of resources**: Various data, services, and content that are accessible through interconnected networks.
  - **A collaboration setup**: Facilitates communication and resource sharing, particularly among research and educational communities globally.

Importance of Internet

- **World Wide Web (WWW)**: A system of interlinked hypertext documents accessed via web browsers.
- **File Transfer Protocol (FTP)**: A protocol used to transfer files between computers over the Internet.
- **Electronic Mail (Email)**: A system for exchanging digital messages between users.
- **Internet Relay Chat (IRC)**: A protocol for real-time text messaging and group chats across the Internet.

Intranet



Intranet

- An **Intranet** is a private TCP/IP network within an organization that uses Internet technologies like web servers and browsers for sharing information and collaboration.
- It provides a secure environment for internal communication and access to various resources within the organization.
- **Key features of an intranet include**:
  - **Content Sharing**: Intranets are used to publish company policies, newsletters, product information, technical support, and tutorials, among other internal resources.
  - **Security**: Unlike public websites, access to an intranet is restricted. Only authorized users with the proper permissions and passwords can access the internal resources.
  - **Internal Access**: Intranet users can access data stored on internal web servers through browsers, using the same interface as public websites, but with restricted access for security.

Extranet

- An **Extranet** is an extension of an organization's intranet that provides secure access to external users or businesses, typically over the Internet.
- It allows authorized external parties—such as vendors, suppliers, and business partners—to access specific internal resources and collaborate securely.
- Key features of an extranet include:
  - **Secure Access**: Access is granted through authentication methods like passwords and user IDs, ensuring only authorized users can interact with the network.
  - **Collaboration with External Parties**: Extranets enable businesses to share resources and information, such as catalogs and supply chain data, with external stakeholders in real-time.
  - **Cost Efficiency**: By using the Internet, extranets reduce the need for expensive dedicated lines and lower communication costs (e.g., phone and fax).
  - **Technology Leverage**: Extranets use familiar technologies like web servers

Ethernet

- **Ethernet** is a family of Local Area Network (LAN) technologies commonly understood through the OSI reference model.
- Ethernet technology names typically consist of three parts:
  - **Speed**: The data transfer rate (e.g., 100 Mbps).
  - **Signal Method**: The signaling technique used—**Baseband** or **Broadband**.
  - **Medium**: The physical medium through which the data is transmitted (e.g., Unshielded Twisted Pair (UTP)).

Example: 100BASE-T

Speed: *100 Mbps*

Signal Method: *Baseband*

Medium: *Unshielded Twisted Pair (UTP)*

Baseband Signaling

- Digital signals are transmitted over a single frequency as discrete electrical pulses.
- **Bidirectional**: Baseband systems can send and receive signals simultaneously.
- **Time-Division Multiplexing (TDM)**: Allows multiple channels over a single transmission line.
- **Signal Regeneration**: Repeaters are used to boost signals over long distances to prevent attenuation (weakening).

Example: *Ethernet uses baseband signaling.*

Broadband Signaling

- Signals are transmitted over a range of frequencies as electromagnetic waves.
- **Unidirectional**: A broadband system can only send or receive signals at any one time.
- **Signal Regeneration**: Amplifiers boost signals to extend their travel distance before attenuation.
- **Frequency-Division Multiplexing (FDM)**: Divides the broadband signal into multiple channels for different transmissions.

  ***Example**: Cable television uses broadband signaling, where multiple channels are carried over a coaxial cable.*

The Reference Model: OSI Model

- The OSI(Open System Interconnection) model defines seven layers for network communication, from hardware to applications, from International Organization for Standardization(ISO).
- Each layer handles specific tasks and interacts only with adjacent layers, making data transmission structured and troubleshooting easier.
- Standardized in 1984, the OSI model provides a universal framework for networking, enabling different systems to communicate.
- While modern networks use the TCP/IP model, the OSI model is still essential for understanding and visualizing network operations.

OSI Model

- ➔ **Provides a universal language** for networking, enabling communication between diverse devices and software.
- ➔ **Simplifies troubleshooting** by isolating issues to specific layers.
- ➔ **Supports faster innovation** with a modular structure, allowing teams to improve individual layers independently.
- ➔ **Ensures compatibility** by enabling seamless integration of new technologies without disrupting the network.

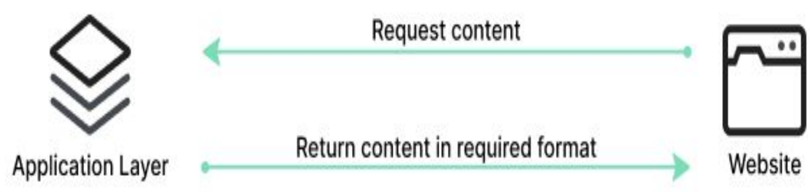
OSI Model

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that datas In a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and Is responsible for controlling ports and sessions
4	Transport Layer	Transmit data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transits raw bit stream over the physical medium

OSI Model: Application Layer

➔ Interface for end-user applications to access network services.

Examples: HTTP (web), FTP (file transfer), SMTP (email), DNS (domain name resolution).



OSI Model: Presentation Layer

➔ Translates data between application and network formats (e.g., ASCII to EBCDIC).

➔ Handles encryption, compression, and data formatting for interoperability.

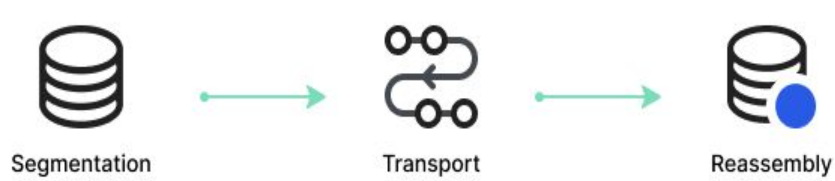
OSI Model: Session Layer

- ➔ Manages session creation, maintenance, and termination.
- ➔ Provides checkpointing and recovery for reliable communication.

OSI Model: Transport Layer

- ➔ Ensures reliable data transfer with error detection and flow control.

  Protocols: TCP (reliable) and UDP (fast but less reliable).



OSI Model: Network Layer

➔ Handles data routing, forwarding, and logical addressing (IP).

Protocols: IP (addressing), ICMP (error reporting), RIP (routing)

OSI Model: Data Link Layer

➔ Ensures node-to-node data transfer and manages MAC addresses.

Technologies: Ethernet (LAN), PPP (node connections).

OSI Model: Physical Layer

➔ Defines the physical connection (cables, switches) and data transmission.

Examples: Fiber Optics, Wi-Fi, and electrical/optical specifications.

Sending an Email

➔ **Application Layer:** The email client (e.g., Outlook) uses SMTP to send the message.

➔ **Presentation Layer:** Formats and encrypts the email for secure transmission.

➔ **Session Layer:** Establishes and manages the session between sender and receiver servers.

➔ **Transport Layer:** Divides the email into packets; TCP ensures reliable delivery.

➔ **Network Layer:** Adds source and destination IP addresses for routing.

➔ **Data Link Layer:** Uses MAC addresses to transmit packets across local networks and correct errors.

➔ **Physical Layer:** Converts data to electrical signals for transmission over cables.

Reversal on Receipt: The layers reverse the process, reassembling, decrypting, and delivering the email to the recipient's inbox.

Advantages of OSI Model

For Network Users and Operators:

- Identifies necessary hardware and software for building networks.
- Simplifies troubleshooting by isolating issues to specific layers.

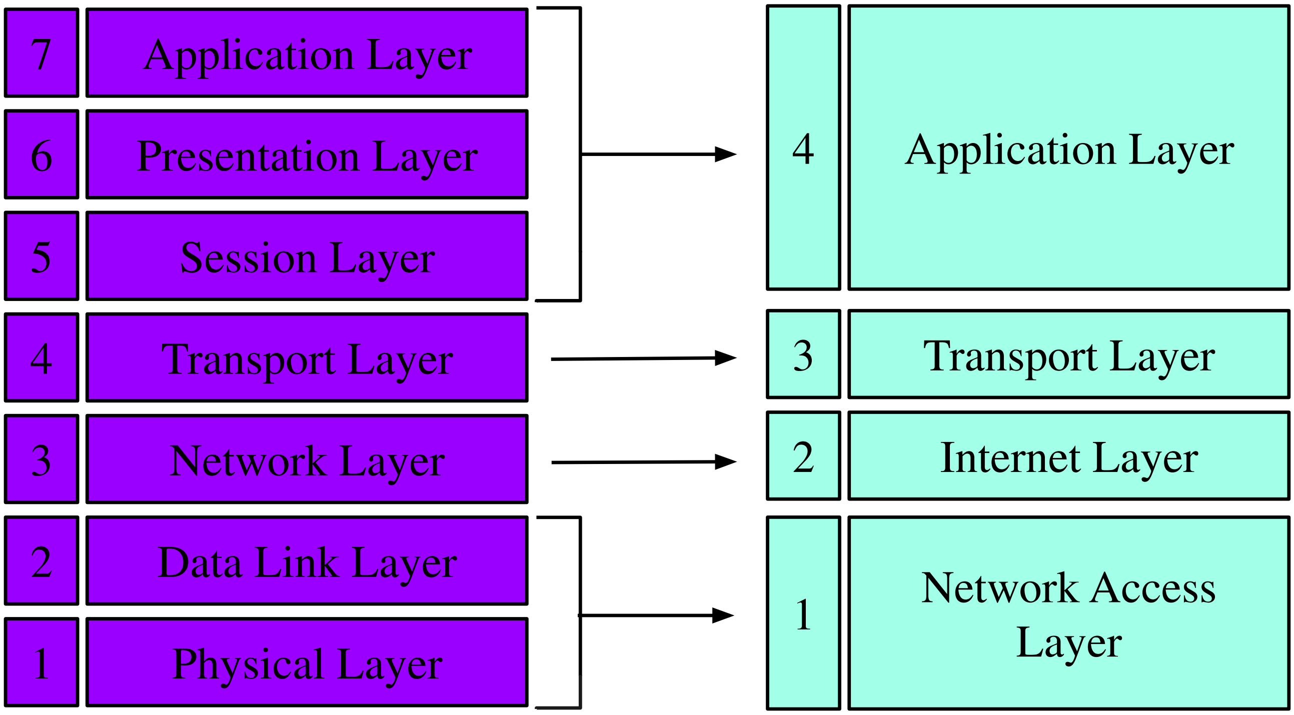
For Device and Software Vendors:

- Enables interoperability between products from different vendors.
- Clarifies which network layers their products support or operate on.

Advantages of OSI Model

- The TCP/IP model is a fundamental framework for computer networking, designed to ensure reliable communication between devices.
- Developed by the Department of Defense in the 1960s, it consists of four layers: the Link Layer, the Internet Layer, the Transport Layer, and the Application Layer.
- Each layer has specific functions that help manage different aspects of network communication.

OSI → TCP/IP



TCP/IP

4	Application Layer	It providing user services or exchanging application data over the network
3	Transport Layer	It includes error checking and recovery mechanisms, flow control, and congestion control
2	Internet Layer	It ensures packets reach their destination by using IP addresses
1	Network Access Layer	It handles the transmission and reception of frames over the physical network medium

TCP/IP

- **Application Layer:** This layer includes protocols used for providing user services or exchanging application data over the network. It supports various applications such as HTTP, FTP, SMTP, and DHCP.
- **Transport Layer:** This layer manages data transmission between devices, ensuring data integrity and order. It includes error checking and recovery mechanisms, flow control, and congestion control.
- **Internet Layer:** This layer provides addressing and routing of packets across networks. It ensures packets reach their destination by using IP addresses.
- **Network Access Layer:** This layer is responsible for moving packets between the internet layer interfaces of two different hosts on the same link. It handles the transmission and reception of frames over the physical network medium.

OSI vs TCP/IP

OSI Model	TCP/IP Model
It is developed by ISO (International Standard Organization)	It is developed by ARPANET (Advanced Research Project Agency Network).
OSI model provides a clear distinction between interfaces, services, and protocols.	TCP/IP doesn't have any clear distinguishing points between services, interfaces, and protocols.
OSI refers to Open Systems Interconnection, consists of seven layers, and Vertical Approach.	TCP refers to Transmission Control Protocol, consists of four layers, and Horizontal Approach.
OSI uses the network layer to define routing standards and protocols.	TCP/IP uses only the Internet layer.

OSI vs TCP/IP

OSI Model	TCP/IP Model
In the OSI model, the transport layer is only connection-oriented.	A layer of the TCP/IP model is both connection-oriented and connectionless.
In the OSI model, the data link layer and physical are separate layers.	In TCP, physical and data link are both combined as a single host-to-network layer.
Session and presentation layers are a part of the OSI model.	There is no session and presentation layer in the TCP model.
The minimum size of the OSI header is 5 bytes.	The minimum header size is 20 bytes.

Networking Protocol

➔ Rules are defined for every step and process at the time of communication among two or more computers.

➔ Networks are needed to follow these protocols to transmit the data successfully.

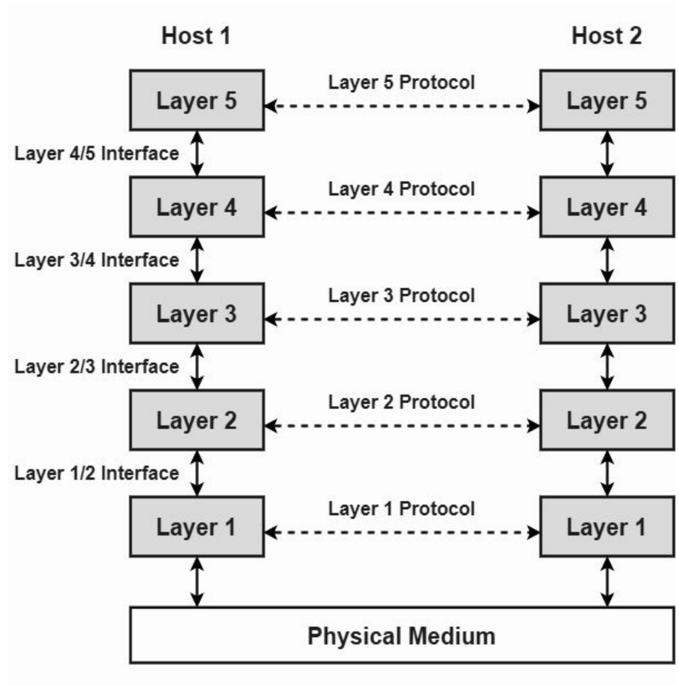
➔ A protocol is a set of rules that governs data communication between computers.

➔ It ensures successful data transmission by defining:

- **Syntax:** Format of transmitted data
- **Semantics:** Meaning of each data segment
- **Timing:** When and how fast data is transmitted

Protocol Hierarchies

- ➔ Networks are structured in layers of hardware and software to simplify design and reduce complexity.
- ➔ Each layer performs specific tasks and provides services to the layer above it, enabling organized communication across networks.



Protocol Hierarchy

Advantages of Protocol Hierarchy

- The layers generally reduce complexity of communication between networks
- It increases network lifetime.
- It also uses energy efficiently.
- It does not require overall knowledge and understanding of network.

Disadvantages of Protocol Hierarchy

- Protocol Hierarchy require a deep understanding of each layers of OSI model.
- Implementation of protocol hierarchy is very costly.
- Every layer in protocol hierarchy introduce overheating in terms of memory, bandwidth and processing.
- Protocol Hierarchy is not scalable for complex networks.

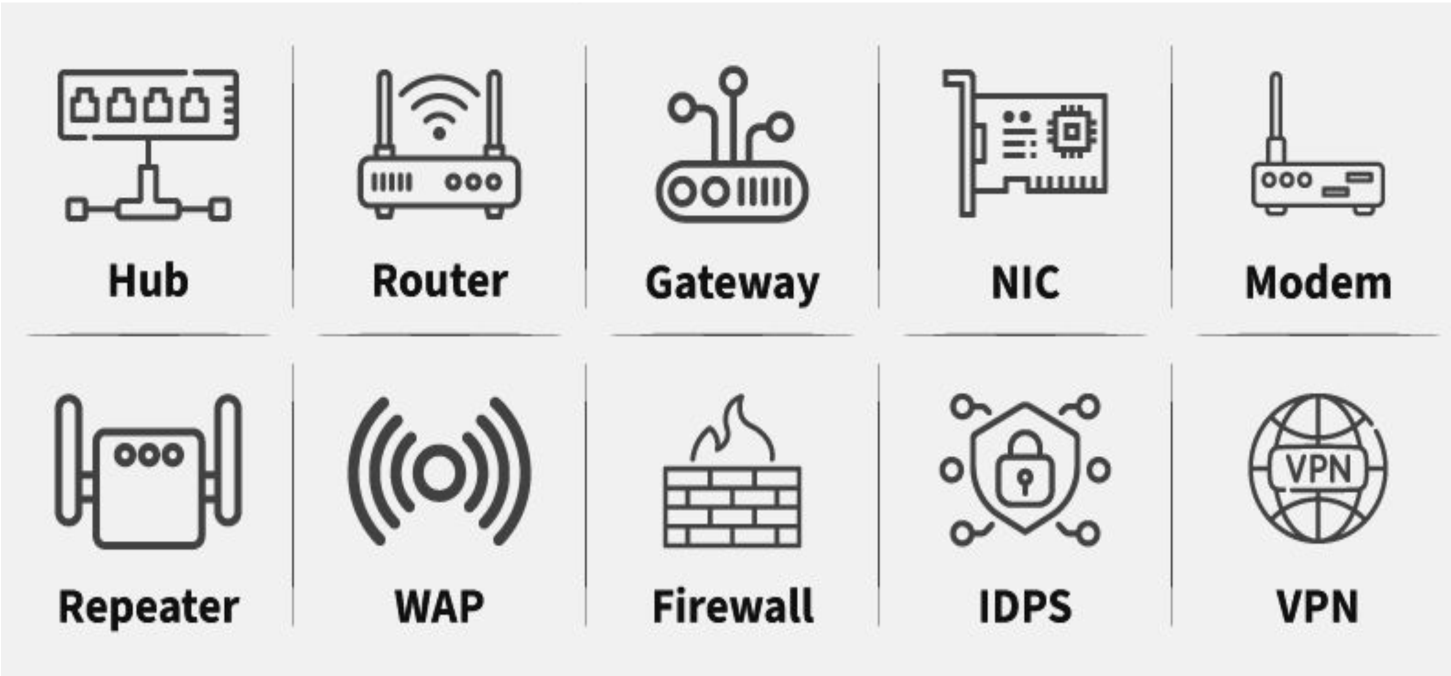
Networking Devices

➔ Network devices are physical devices that allow hardware on a computer network to communicate and interact with each other.
➔ Network devices like hubs, repeaters, bridges, switches, routers, gateways, and brouters help manage and direct data flow in a network.
➔ They ensure efficient communication between connected devices by controlling data transfer, boosting signals, and linking different networks.
➔ Each device serves a specific role, from simple data forwarding to complex routing between networks.

Functions of Networking Devices

➔ Network devices help to send and receive data between different devices.
➔ Network devices allow devices to connect to the network efficiently and securely.
➔ Network devices Improve network speed and manage data flow better.
➔ It protect the network by controlling access and preventing threats.
➔ Expand the network range and solve signal problems.

Networking Devices



Repeater

- A repeater operates at the physical layer.
- Its main function is to amplify (i.e., regenerate) the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network.
- When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength.
- It is a 2-port device.

Hub

- ➔ A hub is a multiport repeater.
- ➔ A hub connects multiple wires coming from different branches.

  For example, the connector in star topology which connects different stations.
- ➔ Hubs cannot filter data, so data packets are sent to all connected devices.
- ➔ They do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Bridge

- ➔ A bridge operates at the data link layer.
- ➔ A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination.
- ➔ It is also used for interconnecting two LANs working on the same protocol.
- ➔ It typically connects multiple network segments and each port is connected to different segment.
- ➔ A bridge is not strictly limited to two ports, it can have multiple ports to connect and manage multiple network segments.

Switch

- ➔ A switch is a multiport bridge with a buffer and a design that can boost its efficiency(a large number of ports imply less traffic) and performance.
- ➔ A switch is a data link layer device.
- ➔ The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only.

Router

- ➔ A router is a device like a switch that routes data packets based on their IP addresses.
- ➔ The router is mainly a Network Layer device.
- ➔ Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets.
- ➔ The router divides the broadcast domains of hosts connected through it.

Gateway

- ➔ A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models.
- ➔ They work as messenger agents that take data from one system, interpret it, and transfer it to another system.
- ➔ Gateways are also called protocol converters and can operate at any network layer.
- ➔ Gateways are generally more complex than switches or routers.

Brouter

- ➔ It is also known as the bridging router is a device that combines features of both bridge and router.
- ➔ It can work either at the data link layer or a network layer.
- ➔ Working as a router, it is capable of routing packets across networks and working as the bridge, it is capable of filtering local area network traffic.

NIC: Network Interface Card

- NIC or network interface card is a network adapter that is used to connect the computer to the network.
- It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it.
- The cable acts as an interface between the computer and the router or modem.